

Description Logics

Volker Haarslev

Concordia University, Computer Science Department
1455 de Maisonneuve Blvd. W.
Montreal, Quebec H3G 1M8, Canada

Description Logics: Introduction



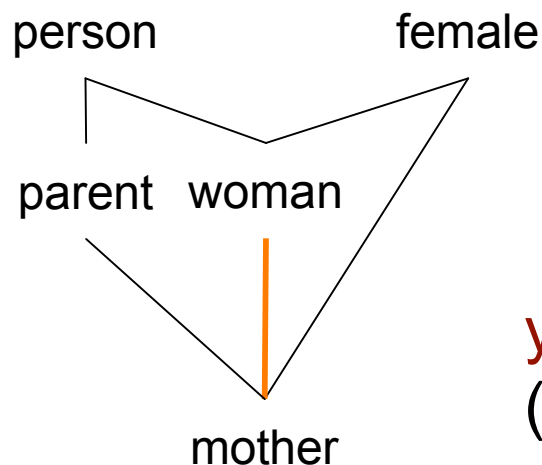
- Important syntactic feature: variable-free notation
 - constructors: $\sqcap$, $\sqcup$, $\neg$, $\exists$, $\forall$
 - standard description logic *ALC*
- Description of concept **parent**
 - $\text{parent} \equiv \text{person} \sqcap \exists \text{has_child}.\text{person}$
- We add two concepts
 - $\text{woman} \equiv \text{female} \sqcap \text{person}$
 - $\text{mother} \equiv \text{female} \sqcap \text{parent}$
- What type of inferences are interesting?
 - **satisfiability** of (named) concepts
 - **subsumption** of (named) concepts

Inference Service: Concept Satisfiability

- The concepts **woman**, **mother**, **parent** are satisfiable
- However, the concept $\neg \text{woman} \sqcap \text{mother}$ is unsatisfiable
- Why? We unfold the definition of **woman** and **mother**
 - $\neg \text{woman} \sqcap \text{mother} \equiv$
 - $\neg(\text{female} \sqcap \text{person}) \sqcap \text{female} \sqcap \text{parent} \equiv$
 - $(\neg \text{female} \sqcup \neg \text{person}) \sqcap \text{female} \sqcap \text{parent} \equiv$
 - $(\neg \text{female} \sqcup \neg \text{person}) \sqcap \text{female} \sqcap \text{parent} \equiv$
 - $\neg \text{person} \sqcap \text{female} \sqcap \text{parent} \equiv$
 - $\neg \text{person} \sqcap \text{female} \sqcap \text{person} \sqcap \exists \text{has_child.person} \equiv$
 - $\neg \text{person} \sqcap \text{female} \sqcap \text{person} \sqcap \exists \text{has_child.person}$
 - **Clash**
- The conjunct $\neg \text{woman} \sqcap \text{mother}$ can never be satisfied

Inference Service: Concept Subsumption

- Consider the question "Is a mother always a woman?"
- Does the concept **woman** subsume the concept **mother**?
- Description logic reasoners offer the computation of a **subsumption hierarchy** (taxonomy) of all named concepts



$\text{parent} \equiv \text{person} \sqcap \exists \text{has_child}.\text{person}$
 $\text{woman} \equiv \text{person} \sqcap \text{female}$
 $\text{mother} \equiv \text{parent} \sqcap \text{female}$

yes, **woman** subsumes **mother**
(see also proof on previous slide)

Description Logics: Semantics (1)

- Translation to first-order predicate logic usually possible
- Declarative** and **compositional** semantics preferred
- Standard Tarski-style interpretation $\mathcal{I} = (\Delta^{\mathcal{I}}, \cdot^{\mathcal{I}})$

Syntax

A

$\neg C$

$C \sqcap D$

$C \sqcup D$

$\forall R.C$

$\exists R.C$

R

$C \sqsubseteq D$

$C \equiv D$

Semantics

$A^{\mathcal{I}} \subseteq \Delta^{\mathcal{I}}$, A is a concept name

$\Delta^{\mathcal{I}} \setminus C^{\mathcal{I}}$

$C^{\mathcal{I}} \cap D^{\mathcal{I}}$

$C^{\mathcal{I}} \cup D^{\mathcal{I}}$

$\{x \in \Delta^{\mathcal{I}} \mid \forall y: (x,y) \in R^{\mathcal{I}} \Rightarrow y \in C^{\mathcal{I}}\}$

$\{x \in \Delta^{\mathcal{I}} \mid \exists y \in \Delta^{\mathcal{I}} : (x,y) \in R^{\mathcal{I}} \wedge y \in C^{\mathcal{I}}\}$

$R^{\mathcal{I}} \subseteq \Delta^{\mathcal{I}} \times \Delta^{\mathcal{I}}$, R is a role name

$C^{\mathcal{I}} \subseteq D^{\mathcal{I}}$

$C^{\mathcal{I}} = D^{\mathcal{I}}$

Concepts

Roles

Axioms



Description Logics: Concept Examples

- $\text{woman} \equiv \text{person} \sqcap \text{female}$
 - $\text{parent} \equiv \text{person} \sqcap \exists \text{has_child}.\text{person}$
 - $\text{mother} \equiv \text{parent} \sqcap \text{female}$
 - $\text{person} \sqsubseteq \forall \text{has_child}.\text{person}$
 - $\text{mother_having_only_female_kids} \equiv \text{mother} \sqcap \forall \text{has_child}.\text{female}$
 - $\text{mother_having_only_daughters} \equiv \text{woman} \sqcap \text{parent} \sqcap \forall \text{has_child}.\text{woman}$
 - $\text{grandma} \equiv \text{woman} \sqcap \exists \text{has_child}.\text{parent}$
 - $\text{great_grandma} \equiv \text{woman} \sqcap \exists \text{has_child}.\exists \text{has_child}.\text{parent}$
- } equivalent

Description Logics: Concept Examples

- $\text{woman} \equiv \text{person} \sqcap \text{female}$
 - $\text{parent} \equiv \text{person} \sqcap \exists \text{has_child}.\text{person}$
 - $\text{mother} \equiv \text{parent} \sqcap \text{female}$
 - $\text{person} \sqsubseteq \forall \text{has_child}.\text{person}$
 - $\text{mother_having_only_female_kids} \equiv \text{mother} \sqcap \forall \text{has_child}.\text{female}$
 - $\text{mother_having_only_daughters} \equiv \text{woman} \sqcap \text{parent} \sqcap \forall \text{has_child}.\text{woman}$
- } equivalent
- $\text{grandma} \equiv \text{woman} \sqcap \exists \text{has_child}.\text{parent}$
 - $\text{great_grandma} \equiv \text{woman} \sqcap \exists \text{has_child}.\exists \text{has_child}.\text{parent}$



Characteristics of DL Semantics



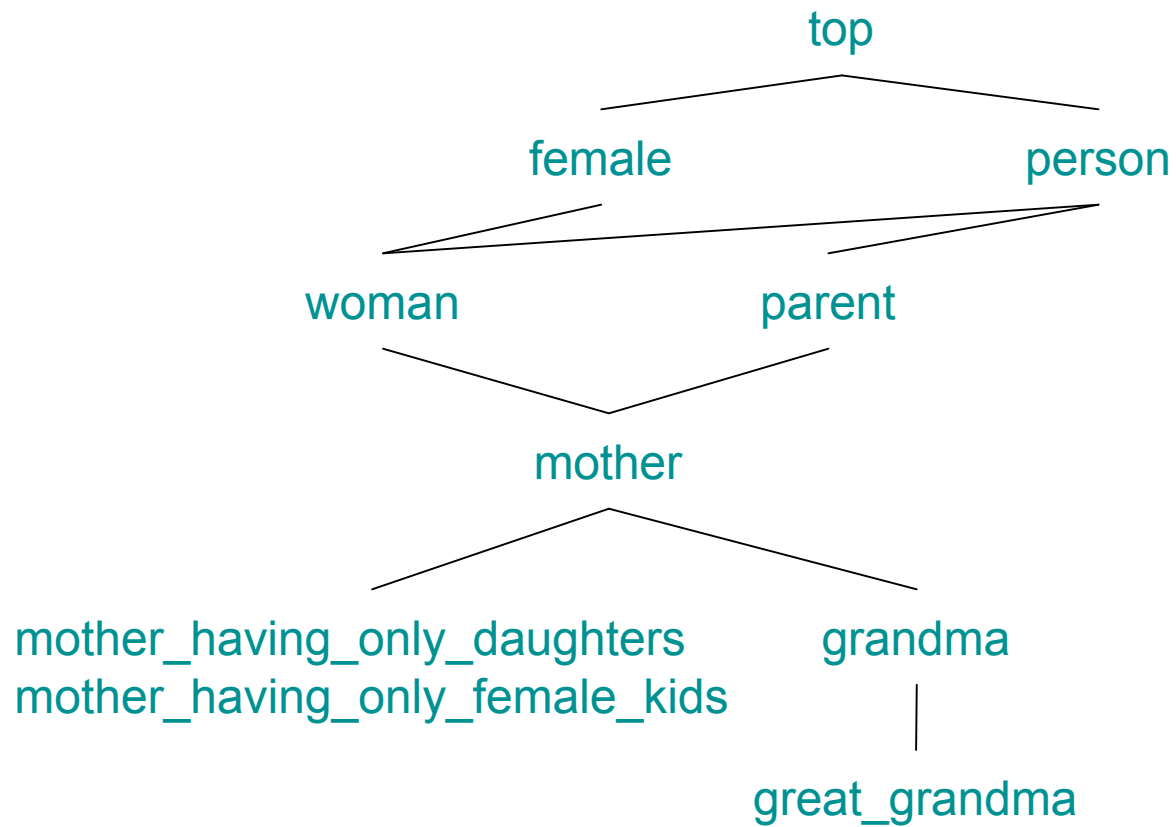
- Interpretation domain can be chosen arbitrarily
- Distinguishing features of description logics
 - domain can be **infinite**
 - **open world assumption**
- A concept **C** is satisfiable iff there exists an interpretation $\mathcal{I}$ such that $C^{\mathcal{I}} \neq \emptyset$
 - $\mathcal{I}$ is called a **model** of **C**
- A concept **C** subsumes a concept **D** iff for all Interpretations $\mathcal{I}$ it holds that $D^{\mathcal{I}} \subseteq C^{\mathcal{I}}$

Description Logics: TBox



- A collection of concept axioms is called a **TBox** (**T**erminological **B**ox)
- Satisfiability of concepts defined w.r.t. a TBox $\mathcal{T}$
- Inference services
 - **TBox coherence**: List all unsatisfiable concept names in $\mathcal{T}$
 - compute **subsumption hierarchy** (taxonomy) of concept names in $\mathcal{T}$
- Why emphasize concept names?
 - ontological decisions of users
 - important concepts will be named

Example Taxonomy



Description Logics: Individuals



- How can we assert knowledge about individuals?
- Assertional axioms
 - concept assertion for an individual a
 - $a:C$ satisfied iff $a^I \in C^I$
 - example: $elizabeth:mother$
 - role assertion for two individuals a and b
 - $(a,b):R$ satisfied iff $(a^I, b^I) \in R^I$
 - example: $(elizabeth, charles):has_child$
- Unique name assumption Not employed per default
 - Different names denote different individuals
 - $a^I \neq b^I$

Description Logics: ABox (1)



- A collection of assertional axioms is called an **ABox** (**A**ssertional **B**ox)
- Satisfiability of assertions defined w.r.t.
 - ABox $\mathcal{A}$
 - TBox $\mathcal{T}$
- Inference services
 - **ABox satisfiability**: Is the collection $\mathcal{A}$ of assertions satisfiable?
 - **Instance checking**: $\text{instance?}(a, C, \mathcal{A})$
Is a an instance of concept C or subsumes C the individual a ?
 - **ABox realization**: compute for all individuals in $\mathcal{A}$ their **most-specific** concept names w.r.t. TBox $\mathcal{T}$

Description Logics: ABox (2)

- New basic inference service: ABox satisfiability
 - $\text{asat}(\mathcal{A})$
- All other inference services can be reduced to asat
 - instance checking:
 $\text{instance?}(a, C, \mathcal{A}) \equiv \neg \text{asat}(\mathcal{A} \cup \{a: \neg C\})$
 - concept satisfiability:
 $\text{sat}(C) \equiv \text{asat}(\{a: C\})$
 - concept subsumption:
 $\text{subsumes}(C, D) \equiv \neg \text{sat}(\neg C \sqcap D) \equiv \neg \text{asat}(\{a: \neg C \sqcap D\})$
- Open world assumption
 - $\mathcal{A} = \{\text{andrew: male}, (\text{charles}, \text{andrew}): \text{has_child}\}$
 - Does $\text{instance?}(\text{charles}, \forall \text{has_child. male}, \mathcal{A})$ hold?

No.
Why?

Description Logics: ABox Example

■ $(\text{male} \sqsubseteq \neg \text{female})$ additional axiom ensuring disjointness

■ $\text{queen_mum} : \text{woman}$

■ $(\text{queen_mum}, \text{elizabeth}) : \text{has_child}$

■ $\text{elizabeth} : \text{woman}$

■ $(\text{elizabeth}, \text{charles}) : \text{has_child}$

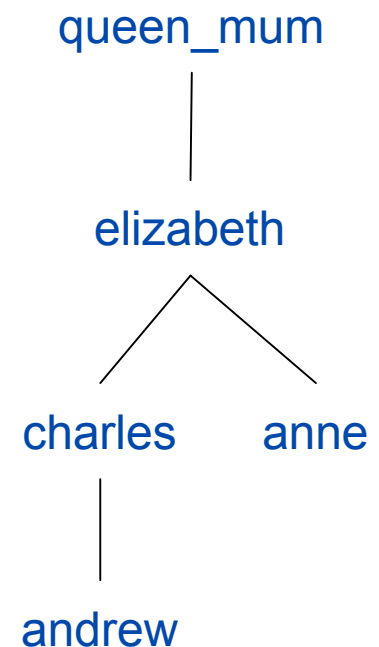
■ $(\text{elizabeth}, \text{anne}) : \text{has_child}$

■ $\text{charles} : \text{parent} \sqcap \text{male}$

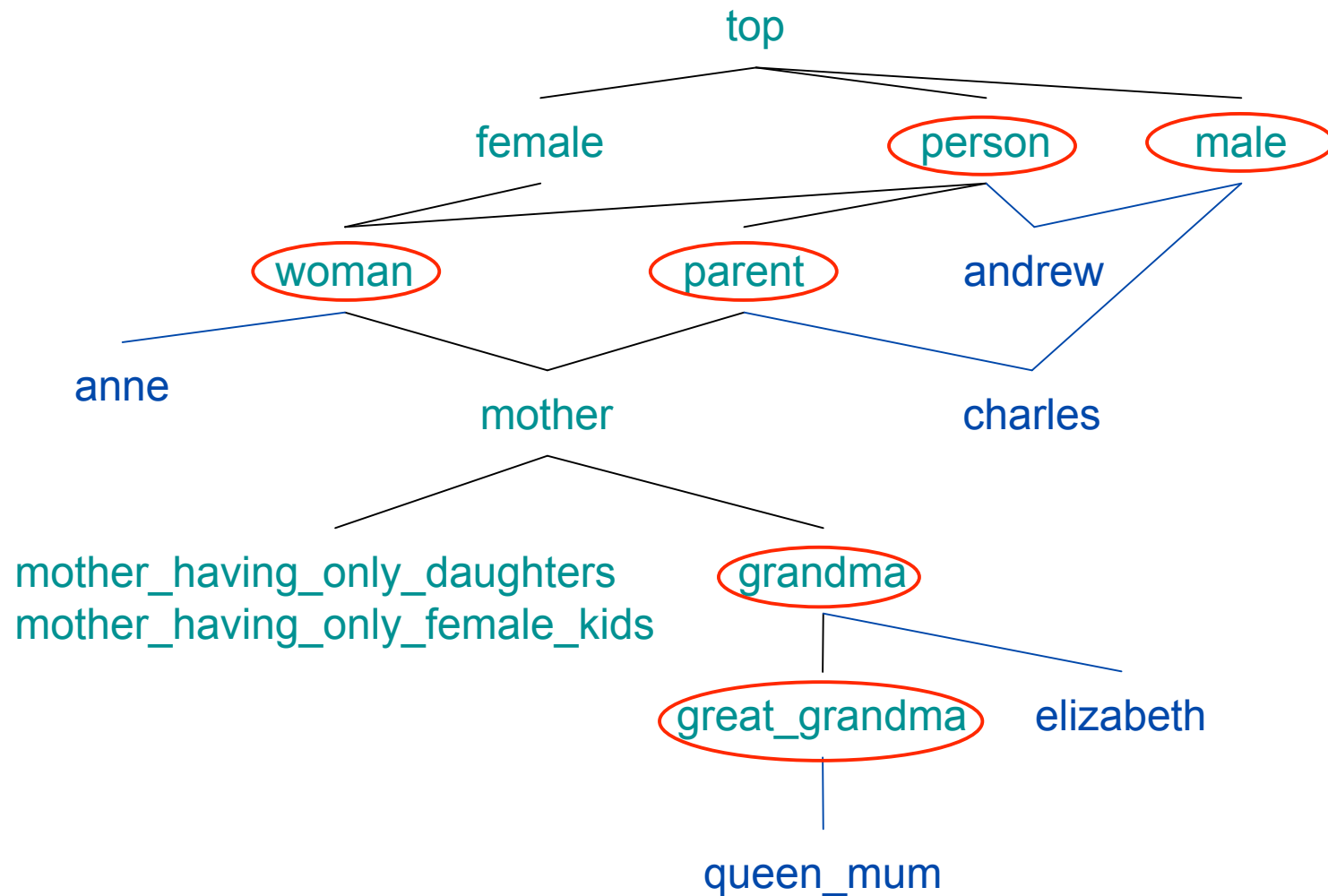
■ $\text{anne} : \text{woman}$

■ $(\text{charles}, \text{andrew}) : \text{has_child}$

■ $\text{andrew} : \text{person} \sqcap \text{male}$



TBox Taxonomy plus Individuals



Open World Assumption



- Can we prove that `instance?(charles,  $\forall$ has_child.male,  $\mathcal{A}$ )` holds?
- No. Although the ABox contains only knowledge about one male child, it is unknown whether additional information about a female child might be added later.
- In order to prevent this, we could add
 - `charles :  $\forall$ has_child.male` or
 - assert that information about a second child will not be added in the future, i.e., **close a role for an individual**
 - Not possible in the logic *ALC* since we need so-called **number restrictions**

More Description Logics Constructors

- Number restrictions on roles ($\mathcal{N}$ resp. $\mathcal{Q}$)
 - simple: $\exists_{\geq 3}\text{has_child}$ or $\exists_{\leq 5}\text{has_child}$
 - qualified: $\exists_{\geq 2}\text{has_child.male}$ or $\exists_{\leq 1}\text{has_child.female}$
- Role hierarchies ($\mathcal{H}$)
 - $\text{has_son} \sqsubseteq \text{has_child}$, $\text{has_daughter} \sqsubseteq \text{has_child}$
 - $\exists_{\geq 2}\text{has_son} \sqcap \exists_{\geq 2}\text{has_daughter} \sqcap \exists_{\leq 4}\text{has_child}$
- Transitive roles ($\mathcal{R}_+$)
 - R declared as transitive:
 - $\text{transitive}(R)$ $R^{\mathcal{I}} = (R^{\mathcal{I}})^+$
 - $\text{transitive}(\text{has_ancestors})$
 $\forall \text{has_ancestors.human}$ applies to all successors of has_ancestors
 - $\text{has_parent} \sqsubseteq \text{has_ancestors}$ demonstrates use of transitive roles in role hierarchies



More Terminological Axioms

- Inverse roles ($\mathcal{I}$):
 - $R \equiv S^{-}$ $(x,y) \in S^{\mathcal{I}} \Rightarrow (y,x) \in R^{\mathcal{I}}$
 - $\text{has_parent} \equiv \text{has_child}^{-}$
- Terminological cycles
 - $\text{human} \sqsubseteq \exists_{\geq 2} \text{has_parent.human}$
 - $\text{binary_tree} \equiv \text{tree} \sqcap \exists_{\leq 2} \text{has_branch} \sqcap \forall \text{has_branch.binary_tree}$
- General (global) axioms
 - axioms that have
 - **not** a concept name on the left-hand side or
 - concept name T (thing, top) as left-hand side
 - **sufficient condition** for concept **grandma**
 $\text{woman} \sqcap \exists \text{has_child}.\exists \text{has_child.person} \sqsubseteq \text{grandma}$
 - **domain** for roles: $\exists \text{has_child.T} \sqsubseteq \text{parent}$
 - **range** for roles: $T \sqsubseteq \forall \text{has_child.person}$